組態
NetApp Solutions

NetApp
May 03, 2022

This PDF was generated from https://docs.netapp.com/zh-tw/netapp-solutions/containers/rh-os-n_use_case_multitenancy_configuration_prerequisites.html on May 03, 2022. Always check docs.netapp.com for the latest.

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組態

對於任何多租戶解決方案、任何使用者都無法存取超過所需的叢集資源。因此、要設定為多租戶組態一部分的一整組資源、會分為叢集管理、儲存管理員和開發人員、分別負責每個專案。

下表概述不同使用者要執行的不同工作：

角色	工作
叢集管理	為不同的應用程式或工作負載建立專案
	為儲存管理員建立Cluster角色 和角色繫結
	建立角色與角色繫結、讓開發人員指派特定專案的存取權
	[選用]設定專案以排程特定節點上的Pod
儲存設備管理	在NetApp ONTAP 上建立SVM
	建立Trident後端
	建立StorageClass
	建立儲存資源配額
開發人員	驗證存取權限、以便在指派的專案中建立或修補PVCS或Pod
	驗證存取權限、以在另一個專案中建立或修補PVCS或Pod
	驗證存取權限、以檢視或編輯專案、資源配額和儲存類別

"下一步：必要條件。"

組態

先決條件

- NetApp ONTAP 產品叢集
- Red Hat OpenShift叢集
- 叢集上安裝的Trident
- 安裝了tridentctl和occ工具並新增至$path的管理工作站
- 管理員存取ONTAP 功能
- 叢集管理存取OpenShift叢集
- 叢集已與Identity Provider整合
- 身分識別供應商的設定可有效區分不同團隊中的使用者

"下一步：叢集管理員工作。"

組態：叢集管理工作

Red Hat OpenShift叢集管理會執行下列工作：

1. 以叢集管理的身分登入Red Hat OpenShift叢集。
2. 建立兩個對應於不同專案的專案。

```
oc create namespace project-1
oc create namespace project-2
```

3. 建立專案1的開發人員角色。

```
cat << EOF | oc create -f -
apiVersion: rbac.authorization.k8s.io/v1
kind: Role
metadata:
  namespace: project-1
  name: developer-project-1
rules:
  - verbs:
      - '*'
    apiGroups:
      - apps
      - batch
      - autoscaling
      - extensions
      - networking.k8s.io
      - policy
      - apps.openshift.io
      - build.openshift.io
      - image.openshift.io
      - ingress.operator.openshift.io
      - route.openshift.io
      - snapshot.storage.k8s.io
      - template.openshift.io
    resources:
      - '*'
  - verbs:
      - '*'
    apiGroups:
      - ''
    resources:
      - bindings
      - configmaps
```

```
      - endpoints
      - events
      - persistentvolumeclaims
      - pods
      - pods/log
      - pods/attach
      - podtemplates
      - replicationcontrollers
      - services
      - limitranges
      - namespaces
      - componentstatuses
      - nodes
  - verbs:
      - '*'
    apiGroups:
      - trident.netapp.io
    resources:
      - tridentsnapshots
EOF
```

本節提供的角色定義只是一個範例。開發人員角色必須根據終端使用者需求加以定義。

1. 同樣地、請為專案2建立開發人員角色。
2. 所有OpenShift和NetApp儲存資源通常由儲存管理員管理。儲存管理員的存取權由安裝Trident時所建立的Trident操作員角色控制。此外、儲存管理員也需要存取資源配額、才能控制儲存設備的使用方式。
3. 在叢集中的所有專案中建立管理資源配額的角色、以將其附加至儲存設備管理員。

```
cat << EOF | oc create -f -
kind: ClusterRole
apiVersion: rbac.authorization.k8s.io/v1
metadata:
  name: resource-quotas-role
rules:
  - verbs:
      - '*'
    apiGroups:
      - ''
    resources:
      - resourcequotas
  - verbs:
      - '*'
    apiGroups:
      - quota.openshift.io
    resources:
      - '*'
EOF
```

4. 請確定叢集已與組織的身分識別提供者整合、而且使用者群組已與叢集群組同步。下列範例顯示身分識別提供者已與叢集整合、並與使用者群組同步。

```
$ oc get groups
NAME                          USERS
ocp-netapp-storage-admins     ocp-netapp-storage-admin
ocp-project-1                 ocp-project-1-user
ocp-project-2                 ocp-project-2-user
```

1. 為儲存管理員設定Cluster勞力 綁定。

```
cat << EOF | oc create -f -
kind: ClusterRoleBinding
apiVersion: rbac.authorization.k8s.io/v1
metadata:
  name: netapp-storage-admin-trident-operator
subjects:
  - kind: Group
    apiGroup: rbac.authorization.k8s.io
    name: ocp-netapp-storage-admins
roleRef:
  apiGroup: rbac.authorization.k8s.io
  kind: ClusterRole
  name: trident-operator
---
kind: ClusterRoleBinding
apiVersion: rbac.authorization.k8s.io/v1
metadata:
  name: netapp-storage-admin-resource-quotas-cr
subjects:
  - kind: Group
    apiGroup: rbac.authorization.k8s.io
    name: ocp-netapp-storage-admins
roleRef:
  apiGroup: rbac.authorization.k8s.io
  kind: ClusterRole
  name: resource-quotas-role
EOF
```

對於儲存管理員、必須綁定兩個角色：Trident運算子和資源配額。

1. 為開發人員建立角色連結、將開發人員專案1角色繫結至專案1中對應的群組（OCP專案-1）。

```
cat << EOF | oc create -f -
kind: RoleBinding
apiVersion: rbac.authorization.k8s.io/v1
metadata:
  name: project-1-developer
  namespace: project-1
subjects:
  - kind: Group
    apiGroup: rbac.authorization.k8s.io
    name: ocp-project-1
roleRef:
  apiGroup: rbac.authorization.k8s.io
  kind: Role
  name: developer-project-1
EOF
```

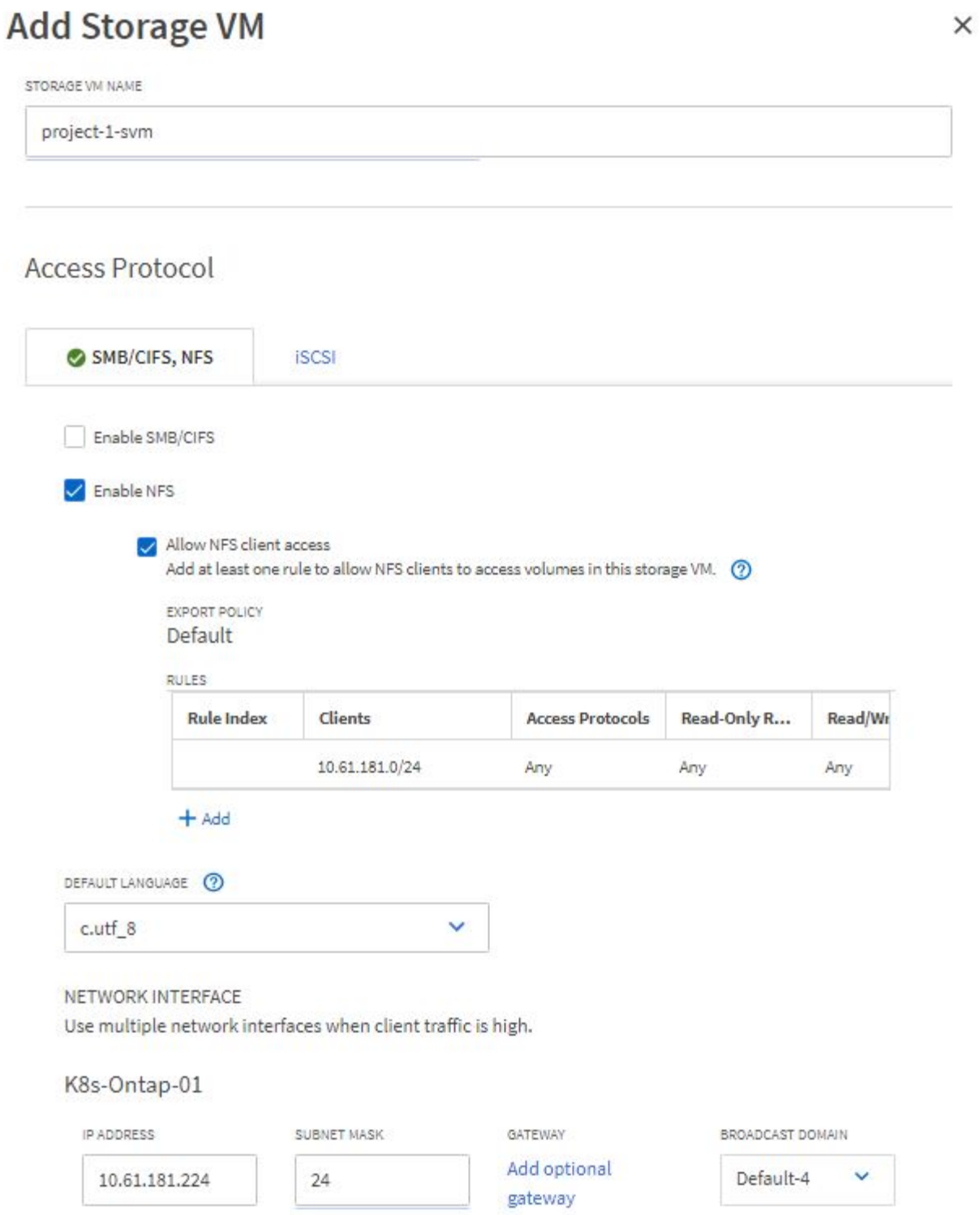
2. 同樣地、請為開發人員建立角色連結至專案2中對應的使用者群組的角色連結。

"下一步：儲存管理員工作。"

組態：儲存設備管理工作

儲存管理員必須設定下列資源：

1. 以admin身分登入NetApp ONTAP 解決方案叢集。
2. 瀏覽至Storage（儲存設備）> Storage VM（儲存設備VM）、然後按一下Add提供所需的詳細資料、建立兩個SVM、一個用於專案1、另一個用於專案2。也可建立vsadmin帳戶來管理SVM及其資源。



1. 以儲存管理員身分登入Red Hat OpenShift叢集。
2. 建立專案1的後端、並將其對應至專案專用的SVM。NetApp建議使用SVM的vsadmin帳戶、將後端連線至SVM、而非ONTAP 使用該叢集管理員。

```
cat << EOF | tridentctl -n trident create backend -f
{
    "version": 1,
    "storageDriverName": "ontap-nas",
    "backendName": "nfs_project_1",
    "managementLIF": "172.21.224.210",
    "dataLIF": "10.61.181.224",
    "svm": "project-1-svm",
    "username": "vsadmin",
    "password": "NetApp123"
}
EOF
```

本例使用的是ONTAP-NAS驅動程式。根據使用案例建立後端時、請使用適當的驅動程式。

我們假設Trident安裝在Trident專案中。

1. 同樣地、請為專案2建立Trident後端、並將其對應至專案2專用的SVM。
2. 接下來、建立儲存類別。建立專案1的儲存類別、並設定storagePools參數、以使用從專屬後端到專案1的儲存資源池。

```
cat << EOF | oc create -f -
apiVersion: storage.k8s.io/v1
kind: StorageClass
metadata:
  name: project-1-sc
provisioner: csi.trident.netapp.io
parameters:
  backendType: ontap-nas
  storagePools: "nfs_project_1:.*"
EOF
```

3. 同樣地、請為專案2建立儲存類別、並將其設定為使用從專屬後端到專案2的儲存資源池。
4. 建立資源配額、以限制專案1中的資源、要求儲存資源來自其他專案專用的儲存設備。

```
cat << EOF | oc create -f -
kind: ResourceQuota
apiVersion: v1
metadata:
  name: project-1-sc-rq
  namespace: project-1
spec:
  hard:
    project-2-sc.storageclass.storage.k8s.io/persistentvolumeclaims: 0
EOF
```

5. 同樣地、也可以建立資源配額、以限制專案2中的資源、要求儲存資源來自其他專案專用的儲存設備。

"下一步：驗證。"

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